$$A = \mathrm{diag}(1, 1, \zeta), \quad \zeta^3 = 1, \quad \zeta \neq 1$$

label j	trace e_1	coefficient e_2	determinant e_3
0	$2 + \zeta$	$1 + 2\zeta$	ζ
1	$-1 + \zeta$	$1 - \zeta$	ζ
2	$-1 - 2\zeta$	$-2 - \zeta$	ζ

First two columns separate. Last column stays fixed.